

Effect of Caffeine on Short-Term Verbal Memory

I. Introduction

Caffeine is widely used for its stimulating effects, particularly among students who believe it enhances alertness and cognitive performance. While its impact on sustained attention has been studied extensively, its effect on memory, especially short-term recall, remains less clear. In this experiment, we tested whether caffeine dosage influences short-term verbal memory by assigning participants to receive one of three doses and measuring their performance on a memory test before and after treatment.

The study employed a blocked design to account for baseline ability, sex, and age, with participants grouped by these variables and randomly assigned to treatment within each block. The primary outcome was score change on a timed verbal memory test. We used randomization-based inference to analyze whether group differences in score change were consistent with a treatment effect. Despite a visible trend of increasing average score with dose, our analysis found no statistically significant effect of caffeine on memory performance. This report presents the design, methodology, results, and interpretation of our findings.

II. Methodology

Our experiment is of a CB[3] design. The treatment for this experiment is a caffeine pill with 3 levels of 0mg (placebo), 100mg, and 200mg. We chose these three levels for ethical reasons; caffeine dosages above 200mg could prove dangerous to health, so we made this our upper limit and spaced out the other two equally. This is a similar constraint that arises if the experiment were to be repeated, since this would mean consuming unsafe levels of caffeine in a short time frame. This ruled out a design such as RM where participants would exceed the 200mg mark. The response measured is the difference in a student's score before and 30 minutes after administration of the treatment on the "Verbal Memory" test from the Human Benchmark website. We think that this response is a reasonable assessment of one's recall ability, which is what we are trying to gauge with our research question. The two factors that we will be accounting for are sex and age. We have two levels of each factor; sex will be classified as either male or female, age will be classified as either 18-20 or 21-23. For age, we chose this division since our units would be college students, which would result in most students being aged 19-22 – our two groups account for an even split of this range. Within every combination of sex and age, we will be blocking on baseline performance, either 'high' or 'low', which will be calculated relative to the median initial scores for subjects in each specific sex/age group. We chose this as the blocking factor because we thought the largest cause of variability in scores

would likely be natural differences in memory. Now, we can be assured that observed differences in outcome are attributed to one of the three factors rather than underlying ability. This means that there will be 8 blocks total. We will be using 48 students from UC Berkeley as experimental and measurement units, which means there will be 6 students per block. Since there are 3 levels of the treatment, two students within a block will receive a unique caffeine dosage.

We deviated slightly from our protocol, as detailed in Appendix C, but the following were our most significant changes. In our protocol, we mentioned that we would have all units complete the study simultaneously. However, due to constraints with the number of students available to participate in our STAT 158 lab section and the classrooms not being able to accommodate over 30 people at once, we were only able to get 24. Having this as our total number of units, however, only achieved a power of ~ 0.25 , which is uncomfortably low. As a result, we decided that we will hold the experiment again for another 24 units the following Monday at 2 PM again with other students we know at UC Berkeley outside of the STAT 158 class. This was meant to minimize variability caused by time of day or day of the week. Additionally, we found it difficult to regroup everyone after the 30-minute period ended, so we sent out an email and made an announcement to retake the ‘Verbal Memory’ portion of the Human Benchmark Test, but there were definitely delays in when people actually decided to take the test. Finally, instead of individually collecting every unit’s sex, age group, and initial performance, we streamlined the process by having all units fill out a Google form where they could input each of the following.

To create blocks, we used a spreadsheet with responses to the Google form and organized people depending on which sex, age group, and baseline performance classification they belonged to. Since two of us were collaborating on this process, with one person working on processing responses from most to least recent, and the other working in the opposite direction, the order in which units were put into blocks was already randomized. We just took the order in which they were listed and assigned them with 0 mg, 100 mg, and 200 mg doses consecutively. The only exception to this was when units reported having already consumed significantly high levels of caffeine throughout the day; we ensured that each of these units was given the placebo. All recorded results were put into table format in Appendix B.

Some shortcomings that we observed in practice were the lack of effort on behalf of certain units or a potential lack of understanding of the assessment. We noticed that certain scores were unusually low (e.g., 1, 3), and this skewed the results of that specific treatment and its corresponding block. Additionally, having to reach our desired unit count of 48 resulted in having to host 2 separate experimental runs. Although we tried to account for confounding variables with the second run on the same day of the week at the same time, there is still a chance that the difference in week caused variances in the data, as well as units being from the general UC Berkeley population instead of our STAT 158 class. As mentioned earlier, there were

also slight delays in some units taking the follow-up assessment after the 30-minute waiting period. With extra time for the caffeine to take effect, this could have also introduced unintended variability into our data.

As for shortcomings in design, we had not initially anticipated having to host two separate experimental trials. This could have also been a factor in our experiment to see if there actually was a significant difference between the two groups, despite our assumption that we controlled for most sources of variability. Additionally, the poor results of certain units make us question the validity of our assessment. We thought that the ‘Verbal Memory’ portion of the Human Benchmark test would be a reasonable gauge of recall, however, the unusually low scores that were observed make us unsure whether these were anomalies or if this was not the best measure that we could have used.

III. Results

We investigated whether short-term verbal memory, as measured by score change on a standardized memory test, differs by caffeine dosage. Given uncertainty about the distributional properties of the response and the modest sample size, we used randomization-based inference for our analysis. This section presents a summary of group-level patterns, formal inference based on permutation tests, and subgroup comparisons across key blocking variables to assess the consistency and robustness of treatment effects.

Mean score changes increased with caffeine dose. Specifically, participants in the 0 mg group improved by 3.6 points (SD = 12.8), those in the 100 mg group improved by 5.7 points (SD = 11.0), and the individuals in the 200 mg group improved by 9.9 points (SD = 14.0).

As shown in Figure 1, score change tended to increase with caffeine dose, but the variability across individuals was substantial. The 0 mg group showed the widest spread, with a median near zero and a range of responses spanning from large gains to sharp declines. The 100 mg group had the most compact distribution, with a consistent upward shift and few extreme values. The 200 mg group had the highest median improvement but also displayed greater variability than the 100 mg group, including one substantial negative outlier. These patterns suggest that while higher doses were associated with greater average gains, individual response variability, especially in the 0 mg and 200 mg groups, complicates the interpretation of dose effects.

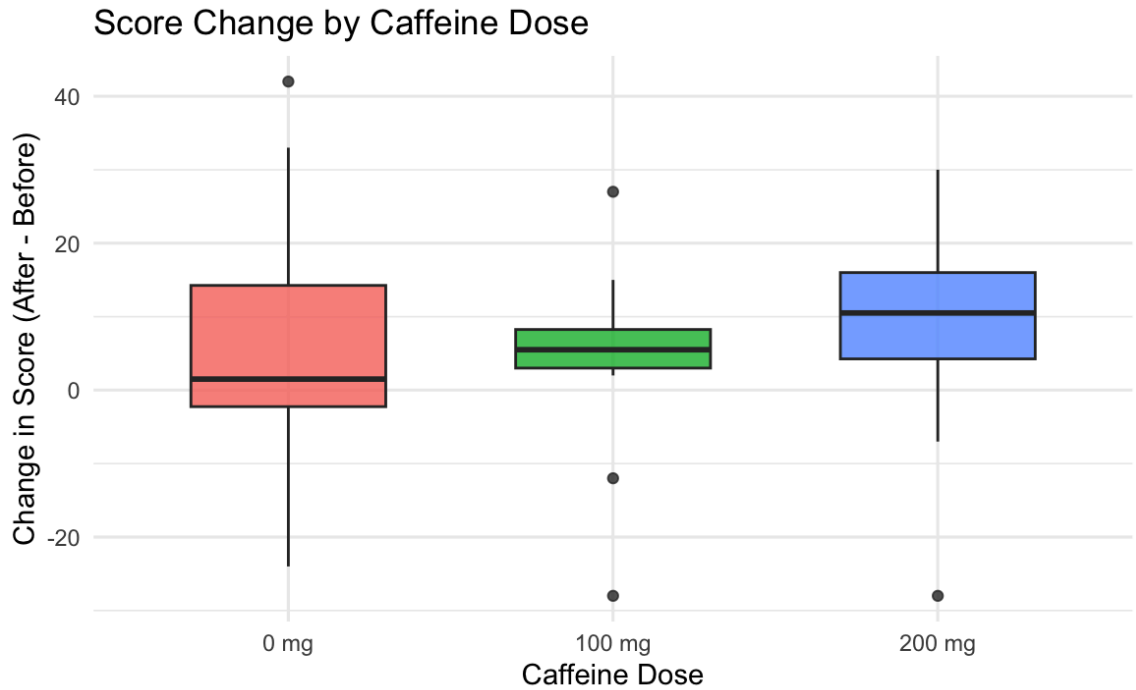


Figure 1. Boxplot of score change by caffeine dose.

To formally assess whether caffeine dosage affected memory performance, we used the F-statistic from a one-way ANOVA on score change as our test statistic. Because the residuals in our data do not clearly satisfy normality or equal-variance assumptions, and because our design included only 16 units per treatment group, we used randomization-based inference rather than relying on model-based p-values. We permuted the treatment labels 5000 times to generate a null distribution of F-statistics under the null hypothesis of no treatment effect. The observed F-statistic was 0.45, and the resulting permutation-based p-value was 0.635 (Figure 2). This suggests that the observed differences in group means are likely attributable to chance and that there is little evidence of a systematic effect of caffeine dose on memory performance in our sample.

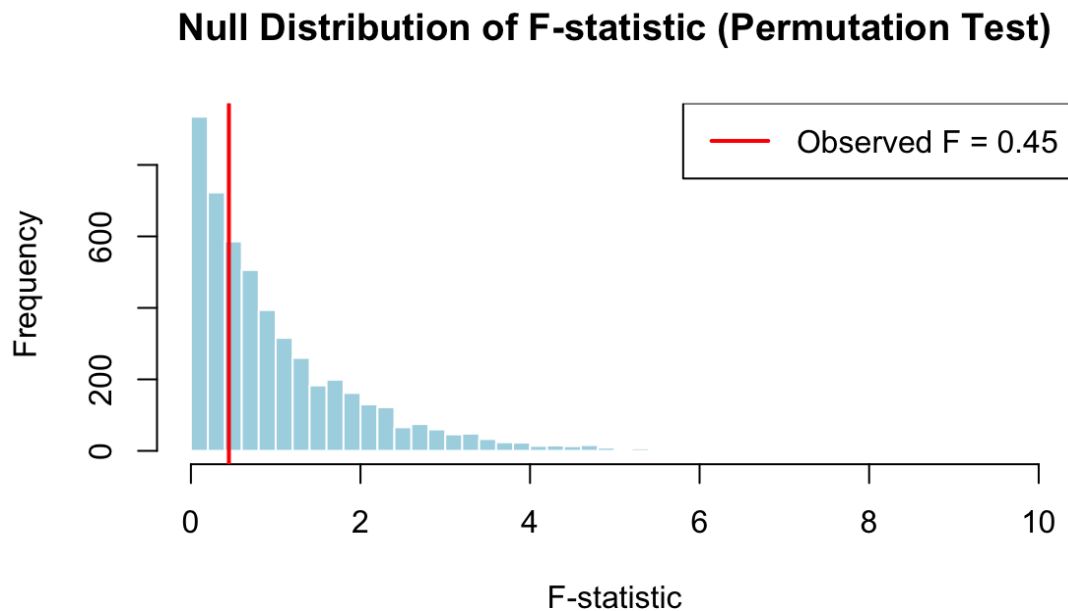


Figure 2. Null distribution of F -statistics from 5000 permutations of treatment labels. The red line indicates the observed F -statistic ($F = 0.45$), which lies well within the body of the null distribution.

We examined whether the effect of caffeine dosage on score change varied across subgroups defined by sex and age. As shown in the interaction plot comparing caffeine dose by sex (Figure 3), the average score change increased steadily for females as the dose of caffeine increased, while males showed a high baseline at 0 mg, a notable drop at 100 mg, and then a recovery at 200 mg. This crossing pattern suggests a potential interaction between sex and dosage, though the differences are not large enough to draw firm conclusions without statistical testing. The age-based interaction plot (Figure 4) shows a similar crossover, as the 18-20 group began higher at 0 mg, dropped at 100 mg, and then sharply increased at 200 mg, whereas the 21-23 group displayed a more linear improvement as the dosage increased. While both plots show some divergence at the middle dose, the dose-response ordering remains broadly consistent across subgroups. These patterns suggest some visual interaction, but do not point to a consistent or reliable moderating effect.

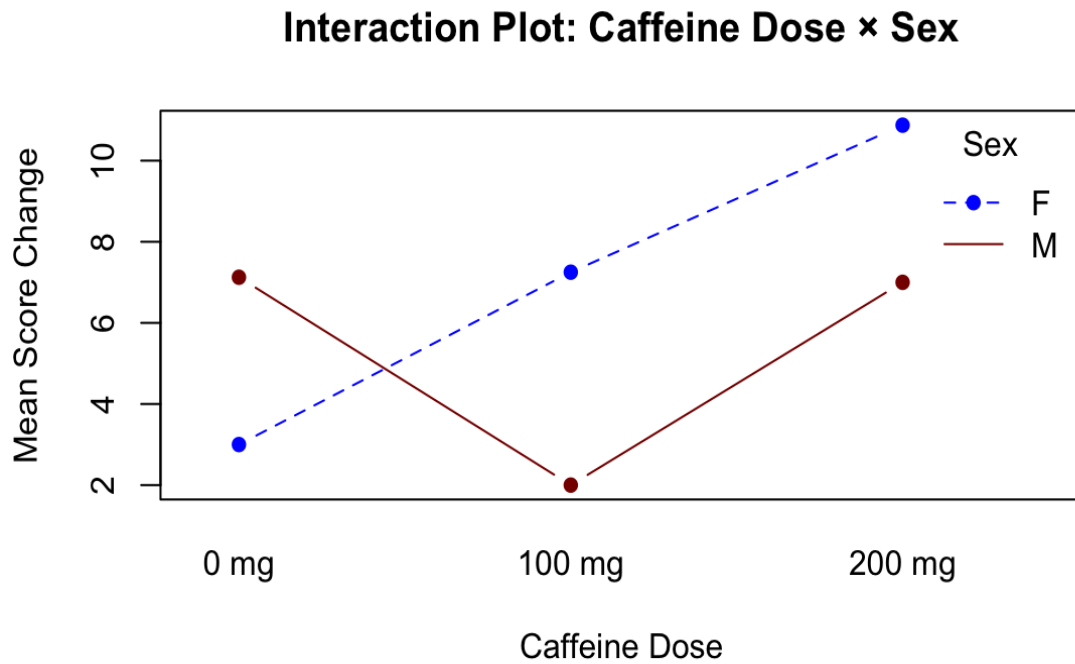


Figure 3. Interaction plot of caffeine dose × sex. Females showed a consistent upward trend in score change, while males dipped at 100 mg before recovering at 200 mg, creating a visible crossover.

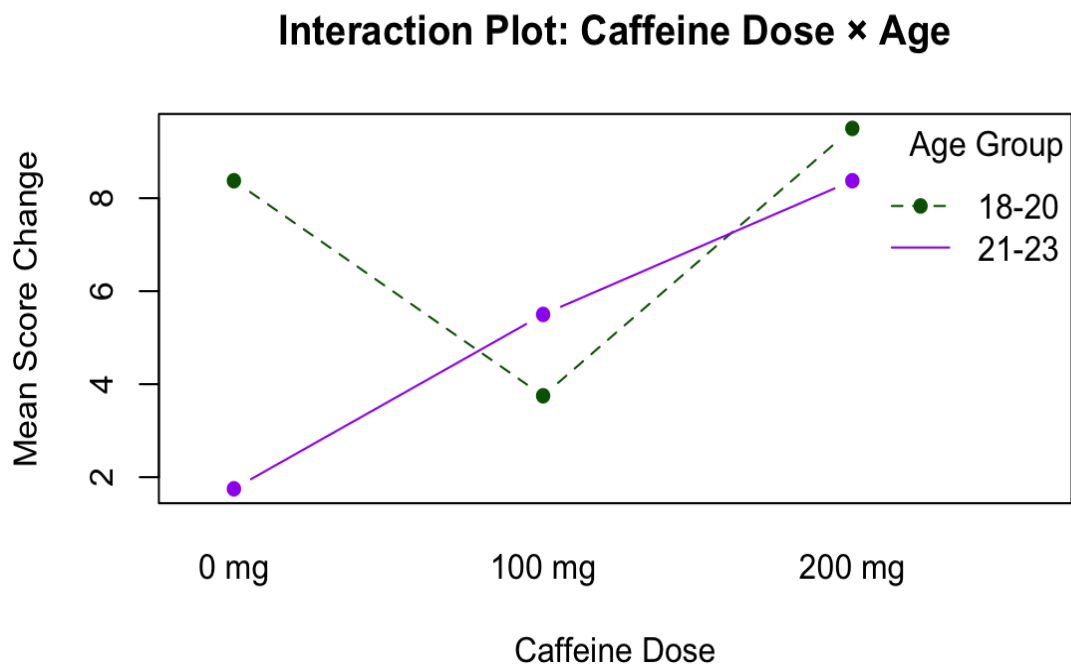


Figure 4. Interaction plot of caffeine dose × age group. The 18–20 group started higher at 0 mg but dropped at 100 mg before rising again, while the 21–23 group showed a steady increase across all doses.

To evaluate whether specific dosage comparisons revealed clearer effects, we performed randomization-based permutation tests for all pairwise contrasts. The largest observed difference was between 200 mg and 100 mg ($\Delta = 4.31$, $p = 0.372$), followed by a 3.88-point increase between 200 mg and 0 mg ($p = 0.489$). The 100 mg vs 0 mg contrast showed almost no difference ($\Delta = -0.44$, $p = 0.936$). While none of these results were statistically significant, the pattern of increasingly higher scores with dosage, particularly at 200 mg, suggests a possible threshold effect that our sample size was underpowered to confirm. These trends, though not definitive, point toward the value of refining the dose range and increasing replication in future experiments to assess whether caffeine meaningfully enhances short-term memory performance.

IV. Discussion

This experiment set out to evaluate whether increasing doses of caffeine lead to improved short-term verbal memory. While average score changes did increase with dosage, randomization-based inference showed no statistically significant treatment effect. Variability within groups, especially in the placebo and 200 mg conditions, was substantial, and treatment differences were well within what could be expected by chance. Visual inspection of interaction plots revealed minor crossover patterns by sex and age, but no consistent subgroup patterns emerged. Together, these findings suggest that caffeine may have a mild effect on memory, but it was not distinguishable from noise in this experimental setting.

The design's strengths included complete blocking and randomization, which helped isolate the treatment effect while controlling for known variation. However, the CB[3] structure limits the degrees of freedom for detecting interactions, and our outcome measure, a single testing moment 30 minutes after ingestion, may not fully capture the dynamics of caffeine's effect. Compliance was also partially self-reported, and caffeine intake prior to the session could not be perfectly controlled. These limitations may have masked true effects or contributed to the variability observed.

Future work could explore a crossover design, allowing for within-subject comparisons and increased sensitivity to detect small effects. Additionally, refining dose levels, like testing 150 mg, or extending the observation window, could better capture peak effects. With caffeine being widely used by students for cognitive enhancement, even small improvements in memory are practically relevant and warrant further investigation under more controlled, targeted conditions.

V. Appendix

Appendix A: Use of AI

We used OpenAI's ChatGPT (GPT-4) solely to assist with formatting our raw data, which was collected via Google Forms, into a structured CSV file for use in R. No AI tools were used in analyzing the data, interpreting results, writing the report, or designing the experiment. All statistical work and narrative content were completed independently by the authors.

Appendix B: Raw Data Table

<u>Unit</u>	<u>Sex</u>	<u>Age</u>	<u>Baseline</u>	<u>Score Before</u>	<u>Score After</u>	<u>Treatment</u>	<u>Notes</u>
1	F	18-20	Low	22	25	0 mg	
2	F	18-20	Low	34	61	100 mg	
3	F	18-20	Low	27	25	200 mg	
4	F	18-20	Low	30	28	0 mg	
5	F	18-20	Low	28	35	100 mg	
6	F	18-20	Low	31	39	200 mg	
7	F	18-20	High	38	21	0 mg	
8	F	18-20	High	44	53	100 mg	
9	F	18-20	High	47	77	200 mg	
10	F	18-20	High	36	39	0 mg	
11	F	18-20	High	35	50	100 mg	
12	F	18-20	High	41	61	200 mg	
13	M	18-20	Low	20	62	0 mg	Had a Celsius today
14	M	18-20	Low	38	50	100 mg	
15	M	18-20	Low	27	60	200 mg	
16	M	18-20	Low	22	25	0 mg	
17	M	18-20	Low	45	50	100 mg	
18	M	18-20	Low	29	35	200 mg	
19	M	18-20	High	52	50	0 mg	

20	M	18-20	High	62	34	100 mg	
21	M	18-20	High	66	80	200 mg	
22	M	18-20	High	75	71	0 mg	
23	M	18-20	High	50	54	100 mg	
24	M	18-20	High	82	54	200 mg	
25	F	21-23	Low	38	53	0 mg	
26	F	21-23	Low	22	24	100 mg	
27	F	21-23	Low	28	49	200 mg	
28	F	21-23	Low	40	48	0 mg	
29	F	21-23	Low	24	30	100 mg	
30	F	21-23	Low	49	58	200 mg	
31	F	21-23	High	72	70	0 mg	
32	F	21-23	High	76	80	100 mg	
33	F	21-23	High	72	78	200 mg	
34	F	21-23	High	65	60	0 mg	
35	F	21-23	High	58	63	100 mg	
36	F	21-23	High	70	77	200 mg	
37	M	21-23	Low	9	24	0 mg	Requested low dose
38	M	21-23	Low	45	48	100 mg	
39	M	21-23	Low	1	3	200 mg	
40	M	21-23	Low	20	24	0 mg	
41	M	21-23	Low	30	42	100 mg	
42	M	21-23	Low	22	35	200 mg	
43	M	21-23	High	50	50	0 mg	
44	M	21-23	High	50	55	100 mg	

45	M	21-23	High	35	50	200 mg	
46	M	21-23	High	40	37	0 mg	
47	M	21-23	High	49	52	100 mg	
48	M	21-23	High	38	43	200 mg	

Appendix C: Protocol Changes

Firstly, there was some confusion in the protocol document on the number of units we would have. We had initially planned on 24, but after our power analysis, we realized that this was far too low. As a result, we had planned on raising this number to 40 since this would be feasible with the number of students who would attend both STAT 158 lab sections. We did not consider, however, the 8 blocks that would become unevenly distributed with 40 units; so, we increased this number to 48 instead. As we had expected, we were unable to seat 48 students in the rooms available to do this experiment concurrently, which resulted in us having to host two separate experimental runs of 24 each (both held on Monday at 2 PM, just one week apart).

As per our initial plan, we were going to simultaneously host all units in one room and have them take the ‘Verbal Memory Test’ on the Human Benchmark website, with instructions presented aloud on how the website works. Due to the chaotic nature of having 24 college students in a room at once, we were unable to make this announcement clearly so that everyone could hear. Additionally, we had not planned on disclosing that there would be a 0 mg (placebo) option, however, many students were hesitant, either given their past caffeine intakes that day or a lack of comfort in taking part in the experiment, leading them to request a lower dosage.

We decided instead of having students individually report their preliminary scores, we could have them indicate their sex, age, and results in a Google form so that we could organize everything in a spreadsheet. This allowed us to organize blocks virtually instead of having students rearrange their seats as we had initially planned; this was simply infeasible due to the already busy nature of the room. We had planned on randomizing treatment assignments within blocks depending on the seating arrangement chosen by students in each block. Instead, we consolidated blocks on our spreadsheet with one person starting from the most recent form responses, and the other starting from the other end. The order in which units appeared in our blocks was randomized in this way, and we just administered the 0 mg, 100 mg, and 200 mg treatments depending on the order in which units were listed.

Finally, we had planned on regrouping all units at the 27-minute mark to have everyone synchronously take the follow-up assessment together. Everyone had already dispersed, so we found it easier to email out a Google form requiring students to report their follow-up scores and made an announcement at the 27- and 30-minute mark. Not forcing students to take it together

introduced slight differences in when they ended up taking it, with some doing it immediately when announced and others completing the form several minutes afterward.

Appendix D: Final R Code

```
#####
# Reading in and formatting data
#####
data <- read.csv("stat158_projectdataset_vF.csv")
data$score_change <- data$Score.After - data$Score.Before

#####
# Boxplot (Figure 1 in text)
#####
library(ggplot2)
ggplot(data, aes(x = Treatment, y = score_change, fill = Treatment)) +
  geom_boxplot(width = 0.6, alpha = 0.8) +
  labs(
    title = "Score Change by Caffeine Dose",
    x = "Caffeine Dose",
    y = "Change in Score (After - Before)"
  ) +
  theme_minimal() +
  theme(legend.position = "none")

#####
# Randomization Test and Permutation Histogram (Figure 2 in text)
#####
observed_anova <- aov(score_change ~ Treatment, data = data)
observed_f <- summary(observed_anova)[[1]]$`F value`[1]
set.seed(158)
n_perm <- 5000
null_f_stats <- replicate(n_perm, {
  shuffled_treatment <- sample(data$Treatment)
  permuted_model <- aov(score_change ~ shuffled_treatment, data = data)
  summary(permuted_model)[[1]]$`F value`[1]
})
p_val <- mean(null_f_stats >= observed_f)
hist(null_f_stats,
     breaks = 40,
     main = "Null Distribution of F-statistic (Permutation Test)",
     xlab = "F-statistic",
     col = "lightblue",
     border = "white")
abline(v = observed_f, col = "red", lwd = 2)
legend("topright", legend = paste("Observed F =", round(observed_f, 2)), col =
"red", lwd = 2)

#####
# Interaction Plot: Caffeine Dose × Sex (Figure 3 in text)
#####
data$Treatment <- factor(data$Treatment, levels = c("0 mg", "100 mg", "200
mg"))
data$Sex <- factor(data$Sex, levels = c("F", "M"))
```

```

interaction.plot(x.factor = data$Treatment,
                trace.factor = data$Sex,
                response = data$score_change,
                fun = mean,
                type = "b",
                legend = TRUE,
                trace.label = "Sex",
                xlab = "Caffeine Dose",
                ylab = "Mean Score Change",
                main = "Interaction Plot: Caffeine Dose × Sex",
                pch = 16,
                col = c("blue", "darkred"))

#####
# Interaction Plot: Caffeine Dose × Age (Figure 4 in text)
#####
data$Age <- factor(data$Age, levels = c("18-20", "21-23"))
interaction.plot(x.factor = data$Treatment,
                trace.factor = data$Age,
                response = data$score_change,
                fun = mean,
                type = "b",
                legend = TRUE,
                trace.label = "Age Group",
                xlab = "Caffeine Dose",
                ylab = "Mean Score Change",
                main = "Interaction Plot: Caffeine Dose × Age",
                pch = 16,
                col = c("darkgreen", "purple"))

#####
# Pairwise Contrasts (100 vs 0, 200 vs 0, 200 vs 100)
#####
run_permutation_test <- function(group1, group2, reps = 5000) {
  subset_data <- subset(data, Treatment %in% c(group1, group2))
  obs_diff <- with(subset_data, mean(score_change[Treatment == group2]) -
                    mean(score_change[Treatment == group1]))

  null_dist <- replicate(reps, {
    shuffled <- sample(subset_data$Treatment)
    mean(subset_data$score_change[shuffled == group2]) -
    mean(subset_data$score_change[shuffled == group1])
  })
  p_val <- mean(abs(null_dist) >= abs(obs_diff))
  list(observed_diff = obs_diff, p_value = p_val, null_distribution =
null_dist)
}

contrast_100_vs_0 <- run_permutation_test("0 mg", "100 mg")
contrast_200_vs_0 <- run_permutation_test("0 mg", "200 mg")
contrast_200_vs_100 <- run_permutation_test("100 mg", "200 mg")
cat("100mg vs 0mg:\n Diff =", contrast_100_vs_0$observed_diff, "\n p =",
contrast_100_vs_0$p_value, "\n\n")
cat("200mg vs 0mg:\n Diff =", contrast_200_vs_0$observed_diff, "\n p =",
contrast_200_vs_0$p_value, "\n\n")
cat("200mg vs 100mg:\n Diff =", contrast_200_vs_100$observed_diff, "\n p =",

```

```
contrast_200_vs_100$p_value, "\n")
```